



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

STATISTICS
PAPER 2

4073/2

2 hours 30 minutes

SPECIMEN PAPER

Additional materials:
Answer booklet
Non-programmable electronic calculator
Mathematical Set
Graph paper

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces at the top of answer booklet provided.

Answer **all** questions in Section A and any **four** questions in Section B

Omission of essential working will result in loss of marks. Decimal answers which are not exact should be given to three significant figures unless stated otherwise.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

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Section A [36 Marks]

Answer **all** questions in this section.

- 1 The table shows the probability distribution for a random variable X .

X	1	2	3	4
$P(X = x)$	0,1	a	0,3	$3a$

Calculate

- (a) the value of a , [2]
- (b) $P(1 < x \leq 4)$ [2]
- 2 (a) Explain the following methods of collecting data.
- (i) observation [1]
- (ii) questionnaire [1]
- (b) State any **one** advantage and any **one** disadvantage of using questionnaire as a method of collecting data. [2]
- 3 A pupil measured the dimensions of the rectangular classroom and got 5 m by 4 m to the nearest metre.
- (a) Find the absolute error in the measurements. [1]
- (b) Find
- (i) the greatest possible area of the room, [2]
- (ii) the least possible perimeter of the room. [2]
- 4 The probability that Munashe is early for school is $\frac{3}{5}$ and Sihle is late for school is $\frac{1}{3}$.
- Find the probability that on a given day
- (a) Munashe is late for school, [1]
- (b) they are both early, [2]
- (c) either Munashe or Sihle is late for school. [2]

- 5 The table shows the prices of fruits per kg in the respective years.

Fruits (1kg)	Price \$ 2008	Price \$ 2025
Oranges	2,00	2,20
Bananas	1,00	0,80

- (a) Using 2008 as the base year, calculate the price relatives for the fruits. [3]
- (b) Hence comment on the results in (a). [2]
- 6 (a) Define the term **statistics**. [1]
- (b) Examine how statistics contributes to
- (i) health care, [2]
- (ii) schools. [2]
- 7 (a) Define the term **bias in sampling**. [1]
- (b) Give any **two** sources of bias. [2]
- (c) Suggest **two** ways of overcoming bias. [2]
- 8 Mr Moyo an agriculture teacher at Magwari Secondary School measured the length, in cm, of millet leaves in the school garden. The results were recorded as follows:
- 10, 9, 8, 10, 11, 12, 10
- (a) Find the
- (i) range of the length of the leaves, [2]
- (ii) interquartile range [3]
- (b) Interpret the value of interquartile range in (a)(ii). [1]
- 9 (a) Explain the following components of a time series:
- (i) irregular or erratic, [2]
- (ii) seasonal variation. [2]
- (b) State the purpose of calculating moving averages in a time series. [2]

10 The following are marks in a Heritage examination:

10 8 7 6 12 14 9 11 13 10

Find the

- (a) mean, [1]
- (b) variance, [3]
- (c) standard deviation and comment. [3]

Section B [48 Marks]

Answer any **four** questions in this section

Each question carries 12 marks.

- 11** Analyse the data given in the table, on prices of products in the years 2023, 2024 and 2025.

Item	Unit	2023	2024	2025
Beef	1 kg	4,00	4,50	3,50
Chicken	1 kg	3,00	3,00	3,10
Goat meat	1 kg	2,00	2,50	3,60

- (a) Define
- (i) base year, [1]
- (ii) price relative. [1]
- (b) Calculate the price relative for beef, chicken and goat meat using 2023 as the base year. [4]
- (c) Comment on the price relative for (b). [3]
- (d) Calculate the unweighted aggregate price index. [2]
- (e) Comment on the results in (d). [1]
- 12** (a) Give the difference between
- (i) population and sample, [2]
- (ii) primary data and secondary data. [2]
- (b) State any **one** source of data for the following categories.
- (i) primary, [1]
- (ii) secondary. [1]
- (c) Give any **two** reasons why sampling is important. [2]
- (d) A village has 150 households with codes numbering 1 to 150. If a sample of 15 households is to be selected for a program on zero tillage, explain how Systematic Sampling could be used to obtain the sample. [4]

- 13 (a) Define the term **sample space** as used in probability. [2]
- (b) A learner is writing a Mathematics test. The probability that she gets the first question correct is $\frac{3}{5}$. If she gets the first question correct the probability that she gets the second question correct is $\frac{3}{4}$. If she gets the first question wrong, the probability that she gets the second question correct is $\frac{1}{4}$.
- (i) Represent the information on a tree diagram. [6]
- (ii) Find the probability that she got both questions correct. [2]
- (iii) Find the probability that she got the second question correct given that the first question was wrong. [2]
- 14 Nhlanhla High School fee collection (X\$1 000) for the whole term for the period 2022 to 2024 is shown in the table.

Year	Term	Total Fees X\$1 000	3 Point Total	3 Point Average
2022	1	68	-	-
	2	52	168	56,0
	3	48	A	57,3
2023	1	72	181	B
	2	61	193	64,3
	3	60	C	68,3
2024	1	84	213	71,0
	2	69	214	D
	3	61	-	-

- (a) Using a scale of 2 cm to represent 1 term on the horizontal axis and 2 cm to represent \$10 000 on the vertical axis, draw a time series graph for the total fees. [4]
- (b) Calculate the missing 3 point moving totals and moving averages A, B, C and D correct to 1 decimal place. [4]
- (c) Plot and join the 3 point moving averages and comment on the trend. [4]

15

Time (minutes)	Frequency
0 – 10	5
10 – 15	12
15 – 20	16
20 – 30	2
30 – 40	10
40 – 50	5

The table shows the time in minutes that each learner in a class took to travel to school on a given day.

- (a) State the modal class. [1]
- (b) Calculate
- (i) the median, [5]
- (ii) the mean and interpret. [6]

- 16 The table below shows prices in dollars of eight items of food purchased from the same supermarket on 1 January 2024 and 1 January 2025.

Item	A	B	C	D	E	F	G	H
2024 Price (x)	66	38	79	53	90	24	32	18
2025 Price (y)	71	42	89	57	95	26	33	19

- (a) Using a scale of 2 cm to represent \$10 on both axis, draw a scatter diagram. [4]
- (b) Calculate $\bar{x}$ and $\bar{y}$ and plot the point $(\bar{x}; \bar{y})$. [3]
- (c) Find and plot the semi-averages. [3]
- (d) Hence or otherwise, draw the line of best fit and comment. [3]